Status	Finished
Started	Wednesday, 21 October 2026, 12:30 PM
Completed	Wednesday, 21 October 2026, 12:39 PM
Duration	8 mins 49 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Extraction
- c. Selection
- d. Distribution

The correct answer is:
Insertion

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are discarded
- c. They are combined to form a single sorted list
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Selection sort
- b. Merge sort
- c. Heap sort
- d. Quick sort

The correct answer is:
Merge sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Decreases the efficiency of algorithms
- c. Makes code execution slower
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. order()
c. sort()
d. arrange()

The correct answer is: sort()

Question **10**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Dynamic programming
c. Backtracking
d. Greedy method

The correct answer is:
Divide and conquer

Question **11**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position
b. Bubbling up the smallest element
c. Sorting the entire list in one pass
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Quick Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Dividing the input into smaller subproblems
b. Sorting the entire list sequentially
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 40,80,90,150,200,300,400,470,70,120
c. 200,470,80,150,40,90,300,400,70,120
d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Sorting

c. Searching

d. Merging

The correct answer is: Sorting